

PINK REPORT

NCCN guidance supports **either BEP×4 or VIP×4 (both Category 1)** for poor-risk disease, explicitly including **mediastinal primary nonseminoma**, with the practical tradeoff that **VIP avoids bleomycin pulmonary toxicity but increases hematologic toxicity and requires routine G-CSF prophylaxis** (NCCN; Nichols; Hinton; de Wit). **Recommendation:** In a bleomycin-risk scenario (e.g., baseline pulmonary vulnerability or strong desire to avoid bleomycin), **VIP×4 is a guideline-endorsed alternative with similar cure/OS in randomized trials**, but plan for **higher myelosuppression/febrile neutropenia risk and mandatory G-CSF**; if BEP is used, follow **structured pulmonary monitoring and early discontinuation thresholds** (NCCN; Nichols; Hinton; Watson; Lauritsen).

References supporting summary ordered by strength of evidence:

Rank	Level of Evidence	Study Title	Authors	Journal (Year)	Notes
1	Guideline	Testicular Cancer	National Comprehensive Cancer Network (NCCN)	NCCN Guidelines (Updated 2025-10-15)	Lists BEP×4 (Category 1, preferred) and VIP×4 (Category 1) for intermediate/poor-risk disease; notes mediastinal primary nonseminoma should be treated with VIP or BEP with careful pulmonary monitoring ; recommends routine G-CSF with VIP . Best-practice guidance for bleomycin use/monitoring (e.g., symptom-triggered evaluation, PFT/DLCO considerations) and mitigation/management of suspected bleomycin lung toxicity.
1	Guideline	Development of a Best-Practice Clinical Guideline for the Use of Bleomycin in the Treatment of Germ Cell Tumours in the UK	Watson RA, De La Peña H, Tsakok MT, et al.	<i>British Journal of Cancer</i> (2018)	

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 1	Guideline	Bleomycin (Prescribing information / label)	U.S. Food and Drug Administration	FDA Label (Updated 2021-12-06)	States bleomycin should be used with extreme caution in patients with compromised pulmonary function ; reinforces pulmonary toxicity risk framing. Meta-analysis of randomized data assessing bleomycin-related pulmonary toxicity risk in first-line regimens; supports quantification of pulmonary risk burden with bleomycin-containing therapy.
 2	Clinical Trial Meta-analysis	Effect of Bleomycin Administration on the Development of Pulmonary Toxicity in Patients With Metastatic Germ Cell Tumors Receiving First-Line Chemotherapy: A Meta-Analysis of Randomized Studies	Necchi A, Miceli R, Oualla K, et al.	<i>Clinical Genitourinary Cancer</i> (2017)	Core randomized comparison of BEP vs VIP in advanced GCT: no significant OS/FFS difference ; establishes comparable efficacy while regimens differ in toxicity profile.
  3	Phase 3 (P3) Trial	Randomized Comparison of Cisplatin and Etoposide and Either Bleomycin or Ifosfamide in Treatment of Advanced Disseminated Germ Cell Tumors (Intergroup/ECOG/SWOG/CALGB)	Nichols CR, Catalano PJ, Crawford ED, et al.	<i>Journal of Clinical Oncology</i> (1998)	Long-term follow-up of intergroup trial: overall survival remains similar between VIP and BEP with extended follow-up; informs durability of cure equivalence.
  3	Phase 3 (P3) Trial	Cisplatin, Etoposide and Either Bleomycin or Ifosfamide in the Treatment of Disseminated Germ Cell Tumors: Final Analysis of an Intergroup Trial	Hinton S, Catalano PJ, Einhorn LH, et al.	<i>Cancer</i> (2003)	Randomized intermediate-prognosis study:
  3	Phase 3 (P3) Trial	Four Cycles of BEP vs Four Cycles of VIP in Intermediate-Prognosis	de Wit R, Stoter G,	<i>British Journal of</i>	

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		Metastatic Testicular Non-Seminoma (EORTC)	Sleijfer DT, et al.	<i>Cancer</i> (1998)	similar disease control ; documents substantially higher leukopenia with VIP vs BEP (hematologic toxicity tradeoff). Not BEP vs VIP, but informs bleomycin administration strategy and pulmonary-toxicity-relevant considerations within BEP-based care pathways.
3	 Phase 3 (P3) Trial	TE-3: 72 h Infusional Versus Bolus Bleomycin in BEP for IGCCCG Good Prognosis Metastatic GCT	Shamash J, Sarker SJ, Huddart R, et al.	<i>Annals of Oncology</i> (2017)	
4	 Phase 2 (P2) Trial				
5	 Phase 1 (P1) Trial				
6	 Large Cohort Study	Extragenadal Germ Cell Tumors of the Mediastinum and Retroperitoneum: Results From an International Analysis	Bokemeyer C, Nichols CR, Droz JP, et al.	<i>Journal of Clinical Oncology</i> (2002)	Large international cohort describing outcomes in extragonadal (incl mediastinal) GCT , supporting historical survival expectations and prognostic context. Cohort focused on primary mediastinal NSGCT outcomes with cisplatin-based therapy; supports disease-specific baseline prognosis context.
7	 Cohort Study	Primary Mediastinal Nonseminomatous Germ Cell Tumors: Results of Modern Therapy Including Cisplatin-Based Chemotherapy	Fizazi K, Culine S, Droz JP, et al.	<i>Journal of Clinical Oncology</i> (1998)	
7	 Cohort Study	Primary Mediastinal Germ Cell Tumors: Real-World Analysis From Two	Gangadhar B, Nagpal C, Sharma A, et al.	<i>JCO Global Oncology</i> (2025)	Contemporary real-world series including patients treated with BEP

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		Tertiary Cancer Centers in India			and VIP ; provides pragmatic outcomes context (heterogeneous histologies; observational). Develops/validates risk factors for bleomycin lung toxicity (e.g., renal function, age, stage, cumulative dose), supporting risk stratification beyond asthma-specific evidence. Observational evaluation of PFT/DLCO changes and pulmonary outcomes in BEP-treated patients; informs monitoring expectations. Small clinical study suggesting atopy may associate with higher bleomycin pulmonary toxicity risk; hypothesis-generating and not asthma-specific proof. Real-world outcomes evaluating bleomycin pneumonitis and G-CSF ; cited to support the clinical question of whether G-CSF use materially worsens
7	Cohort Study	Predicting the Risk of Bleomycin Lung Toxicity in Patients With Germ-Cell Tumours	O'Sullivan JM, Huddart RA, Norman AR, et al.	<i>Annals of Oncology</i> (2003)	
7	Cohort Study	Pulmonary Function in Patients With Germ Cell Cancer Treated With Bleomycin, Etoposide, and Cisplatin	Lauritsen J, Kier MG, Bandak M, et al.	<i>Journal of Clinical Oncology</i> (2016)	
7	Cohort Study	The Role of Atopy in the Pathogenesis of Bleomycin Pulmonary Toxicity	Ozyigit LP, Aktas EC, Senbas ZA, et al.	<i>Respiratory Medicine</i> (2019)	
7	Cohort Study	Decade of insights: Bleomycin-induced pneumonitis and use of G-CSF in chemotherapy-treated GCT and Hodgkin lymphoma—Real-world outcomes	Shah M, Mortazavi A, Folefac E, et al.	<i>Journal of Clinical Oncology</i> (2024; Suppl abstract)	

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 8	Animal Model				pulmonary toxicity risk.